

Inter-VLAN Routing Skills Check

Skills Check Overview

In this performance-based assessment, you will design, build, configure, test, troubleshoot, and document a small business network in Cisco Packet Tracer.

You will demonstrate your ability to:

- Configure VLANs on managed switches.
- Assign switch ports to the correct VLAN.
- Configure trunk links between switches and a router.
- Configure router-on-a-stick inter-VLAN routing.
- Apply IPv4 addressing, subnet masks, and default gateways.
- Verify connectivity using ping tests and Cisco IOS show commands.
- Document a network professionally.

Specific external standards could not be verified for this assessment.

Learning Objectives

By the end of this skills check, you will be able to:

1. Configure VLANs, access ports, and trunk ports on Cisco 2960 switches.
2. Configure router subinterfaces using IEEE 802.1Q encapsulation.
3. Subnet a private IPv4 network and correctly assign addresses to network devices.
4. Explain the purpose of default gateways and Layer 3 forwarding between VLANs.
5. Use ping tests and Cisco IOS show commands to verify and troubleshoot network connectivity.

Scenario

You are a junior network technician for BrightWave Office Solutions, a small business with two departments:

- **Department A:** Administration
- **Department B:** Technical Support

The business wants to separate department traffic for organization and security. Each department must have its own VLAN and IP subnet. Employees in the same department must be able to communicate with one another. The router must provide communication between the two VLANs.

You have been asked to build and document a working prototype of this network in Cisco Packet Tracer.

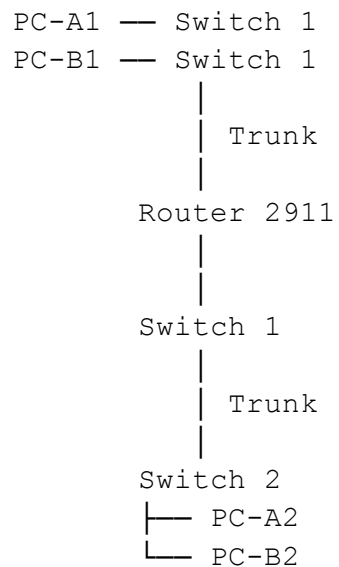
Network Requirements

Build the following topology:

- 1 Cisco 2911 router
- 2 Cisco 2960 switches
- 4 PCs
- 2 PCs per VLAN
- VLAN 10 for Department A
- VLAN 20 for Department B
- One trunk connection from the router to Switch 1
- One trunk connection between Switch 1 and Switch 2
- Access connections from PCs to the switches

Required Topology

Use the following logical arrangement:



Connect the devices using copper straight-through cables unless Packet Tracer requires a different cable type.

Suggested Port Assignments

Device	Port	Connected Device	Port Type	VLAN
Switch 1	Fa0/1	PC-A1	Access	VLAN 10
Switch 1	Fa0/2	PC-B1	Access	VLAN 20
Switch 1	Fa0/23	Switch 2	Trunk	VLANs 10, 20
Switch 1	Gi0/1	Router G0/0	Trunk	VLANs 10, 20
Switch 2	Fa0/1	PC-A2	Access	VLAN 10
Switch 2	Fa0/2	PC-B2	Access	VLAN 20

You may use different ports if you clearly document the changes.

Technical Specifications

VLAN Configuration

Configure the following VLANs on both switches:

VLAN ID	VLAN Name	Department
10	DEPARTMENT_A	Administration
20	DEPARTMENT_B	Technical Support

Required access-port assignments:

- Switch 1 FastEthernet0/1: VLAN 10
- Switch 1 FastEthernet0/2: VLAN 20
- Switch 2 FastEthernet0/1: VLAN 10
- Switch 2 FastEthernet0/2: VLAN 20

Required trunk ports:

- Switch 1 GigabitEthernet0/1 to Router GigabitEthernet0/0
- Switch 1 FastEthernet0/23 to Switch 2 FastEthernet0/23

The trunk links must carry VLAN 10 and VLAN 20 traffic.

IP Addressing Scheme

Use separate private Class C networks for each VLAN: 192.168.10.0/24 for VLAN 10 and 192.168.20.0/24 for VLAN 20.

VLAN	Department	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
10	Department A	192.168.10.0/24	255.255.255.0	192.168.10.1–192.168.10.254	192.168.10.255
20	Department B	192.168.20.0/24	255.255.255.0	192.168.20.1–192.168.20.254	192.168.20.255

Device Addressing Table

Device	Interface	VLAN	IPv4 Address	Subnet Mask	Default Gateway
Router	G0/0.10	10	192.168.10.1	255.255.255.0	Not applicable
Router	G0/0.20	20	192.168.20.1	255.255.255.0	Not applicable
PC-A1	FastEthernet0	10	192.168.10.10	255.255.255.0	192.168.10.1
PC-A2	FastEthernet0	10	192.168.10.11	255.255.255.0	192.168.10.1
PC-B1	FastEthernet0	20	192.168.20.10	255.255.255.0	192.168.20.1
PC-B2	FastEthernet0	20	192.168.20.11	255.255.255.0	192.168.20.1

Do not assign the network address or broadcast address to a PC interface.

Router Configuration

Configure router-on-a-stick using Router GigabitEthernet0/0.

Required subinterfaces:

- G0/0.10 for VLAN 10
- G0/0.20 for VLAN 20
- IEEE 802.1Q encapsulation
- Correct default gateway address for each VLAN
- Physical G0/0 interface enabled

The router should provide Layer 3 forwarding between VLAN 10 and VLAN 20.

Example command structure:

```
enable
configure terminal
interface gigabitEthernet 0/0
no shutdown
exit

interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
end
write memory
```

You must enter and document the commands yourself. Do not copy commands without understanding their purpose.

Switch Configuration

Each switch must include:

- VLAN 10 named DEPARTMENT_A
- VLAN 20 named DEPARTMENT_B
- Correct access-port assignments
- Correct trunk-port configuration
- Active interfaces
- Saved configuration

Example VLAN command structure:

```
enable
configure terminal
vlan 10
name DEPARTMENT_A
exit
```

```
vlan 20
name DEPARTMENT_B
exit
```

Example access-port command structure:

```
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
no shutdown
exit
```

Example trunk-port command structure:

```
interface gigabitEthernet 0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
no shutdown
exit
```

Use the appropriate interface and port numbers for each device.

Step-by-Step Tasks

1. Build the Physical Topology

1. Open Cisco Packet Tracer.
2. Add one Cisco 2911 router.
3. Add two Cisco 2960 switches.
4. Add four PCs.
5. Connect the devices according to the required topology.
6. Label each device clearly:
 - R1
 - S1
 - S2
 - PC-A1
 - PC-A2
 - PC-B1
 - PC-B2

2. Configure VLANs on Both Switches

On Switch 1 and Switch 2:

1. Enter privileged EXEC mode.
2. Enter global configuration mode.
3. Create VLAN 10.
4. Name VLAN 10 `DEPARTMENT_A`.
5. Create VLAN 20.
6. Name VLAN 20 `DEPARTMENT_B`.
7. Exit configuration mode.
8. Save the configuration.

3. Configure Access Ports

Configure the PC-facing ports:

- Department A PCs must use VLAN 10.
- Department B PCs must use VLAN 20.
- Set each PC-facing port to access mode.
- Enable each port.
- Verify the assignments.

4. Configure Trunk Ports

Configure:

- The Switch 1-to-Switch 2 connection as a trunk.
- The Switch 1-to-Router connection as a trunk.
- VLANs 10 and 20 as allowed VLANs.
- Each trunk interface as enabled.

5. Configure Router Subinterfaces

On Router 1:

1. Enable GigabitEthernet0/0.
2. Create subinterface G0/0.10.
3. Assign VLAN 10 encapsulation.
4. Assign the VLAN 10 gateway address.
5. Create subinterface G0/0.20.
6. Assign VLAN 20 encapsulation.
7. Assign the VLAN 20 gateway address.
8. Verify the interfaces are operational.
9. Save the configuration.

6. Configure PC IP Settings

Configure each PC with the correct:

- IPv4 address
- Subnet mask
- Default gateway

Use the addressing table provided in this assessment.

7. Test Connectivity

Perform and record:

1. Ping tests between PCs in VLAN 10.
2. Ping tests between PCs in VLAN 20.
3. Ping tests from VLAN 10 to VLAN 20.
4. Ping tests from each PC to its default gateway.
5. Verification commands on both switches and the router.

8. Document the Configuration

Create a professional documentation file containing:

- Network diagram
- IP addressing table
- Configuration command log
- Verification screenshots
- Connectivity results
- Troubleshooting log
- Brief explanation of how inter-VLAN routing works in this topology

Testing and Verification Requirements

Record the date and time of each major verification activity.

Required Ping Tests

Test	Expected Result
PC-A1 to PC-A2	Successful
PC-B1 to PC-B2	Successful
PC-A1 to 192.168.10.1	Successful
PC-A2 to 192.168.10.1	Successful
PC-B1 to 192.168.20.1	Successful
PC-B2 to 192.168.20.1	Successful
PC-A1 to PC-B1	Successful
PC-A2 to PC-B2	Successful
PC-A1 to PC-B2	Successful
PC-B1 to PC-A2	Successful

A successful test should show replies with no packet loss. If a test fails, record the failure and troubleshooting steps.

Required Switch Commands

Run and capture the output of the following commands on both switches:

```
show vlan brief
show interfaces trunk
show interfaces status
show running-config
show mac address-table
```

Use the results to verify:

- VLANs exist and have correct names.
- Access ports are assigned to the correct VLAN.
- Trunk links are active.
- MAC addresses are being learned.
- Configuration is present in the running configuration.

Required Router Commands

Run and capture the output of the following commands on the router:

```
show ip interface brief
show running-config
show interfaces
show ip route
```

Use the results to verify:

- G0/0 is enabled.
- Both subinterfaces exist.
- Both subinterfaces have correct IP addresses.
- Both subinterfaces are operational.
- Connected routes for both VLAN subnets appear in the routing table.

Expected Router Routing Table Entries

The routing table should show connected networks for:

- 192.168.10.0/24
- 192.168.20.0/24

Optional PC Verification

From each PC, use the Desktop tab to verify IP Configuration and use the Command Prompt to run:

```
ipconfig
ping <destination IP address>
arp -a
```

The ARP table should eventually show MAC address information for devices reached on the local network.

Documentation Requirements

Submit one professional documentation file in PDF or Word format.

Required Documentation Components

1. Network Diagram

Include a diagram showing:

- Router
- Both switches
- All four PCs
- Device names
- Interface names
- Trunk links
- VLAN assignments
- IP addresses where appropriate

The diagram may be created in Packet Tracer or drawn using another tool.

2. IP Addressing Table

Include a complete table with:

- Device name
- Interface
- VLAN
- IPv4 address
- Subnet mask
- Default gateway
- Notes

3. Configuration Commands Log

Include all commands entered on:

- Router 1
- Switch 1
- Switch 2

Organize the commands by device and configuration purpose.

4. Verification Screenshots

Include readable screenshots showing:

- `show vlan brief`
- `show interfaces trunk`
- `show ip interface brief`
- `show ip route`
- Relevant portions of `show running-config`
- Successful ping results

Screenshots must include enough of the Packet Tracer window or CLI prompt to identify the device.

5. Connectivity Test Results

Include a table with:

- Source device
- Destination device or address
- Test type
- Expected result
- Actual result
- Pass or fail
- Notes

6. Troubleshooting Log

For every issue encountered, record:

- Date and time
- Device or interface affected
- Observed symptom
- Possible cause
- Troubleshooting commands or actions
- Resolution
- Final verification result

If no issues occur, write: “No troubleshooting issues encountered during configuration.”

Professional Standards

Your documentation must demonstrate:

- Clear labeling and organization.
- Correct technical terminology.
- Complete command syntax.
- Consistent device and interface names.
- Readable screenshots.
- Accurate IP addressing information.
- Timestamped entries where appropriate.
- Complete sentences in explanations and troubleshooting notes.
- A logical sequence that another technician could follow.

Submission Requirements

Submit the following files:

1. Cisco Packet Tracer file:

- o .pkt format

2. Documentation:

- o .pdf or .docx format

File Naming Convention

Use the following format:

```
LastName_FirstName_InterVLAN_SkillsCheck.pkt  
LastName_FirstName_InterVLAN_Documentation.pdf
```

Example:

```
Garcia_Maya_InterVLAN_SkillsCheck.pkt  
Garcia_Maya_InterVLAN_Documentation.pdf
```

Submission Checklist

Before submitting, verify that:

- All required devices are present.
- VLANs are configured on both switches.
- Access ports are assigned correctly.
- Trunk links are operational.
- Router subinterfaces are configured.
- All PC addresses are correct.
- All required pings are successful.
- Configurations are saved.
- Documentation is complete.
- Both files use the required naming convention.

Troubleshooting Tips

Trunk Port Not Configured

Symptoms:

- Devices on the same VLAN but different switches cannot communicate.
- `show interfaces trunk` does not display the expected interface.
- VLANs do not appear to pass between switches.

Check:

```
show interfaces trunk
show running-config
```

Verify that both ends of the link are configured as trunks and that VLANs 10 and 20 are allowed.

Incorrect Subnet Mask

Symptoms:

- A PC cannot reach its default gateway.
- A PC is placed in the wrong logical network.
- Some pings work while others fail.

Check:

- PC IP address
- Subnet mask
- Router subinterface mask
- Network and broadcast addresses

VLAN 10 uses the 192.168.10.0/24 network and VLAN 20 uses the 192.168.20.0/24 network.

Wrong Default Gateway

Symptoms:

- A PC can communicate with some local devices but cannot reach the other VLAN.
- Pings to the router gateway fail or inter-VLAN pings fail.

Check:

- VLAN 10 PCs must use 192.168.10.1.
- VLAN 20 PCs must use 192.168.20.1.

Encapsulation Mismatch

Symptoms:

- Router subinterfaces exist, but inter-VLAN communication fails.
- The router does not properly process tagged frames.

Check:

```
show running-config
show ip interface brief
```

Verify:

```
encapsulation dot1Q 10
encapsulation dot1Q 20
```

The VLAN number in the encapsulation command must match the VLAN being routed.

Interface Shutdown

Symptoms:

- An interface displays as administratively down.
- Link indicators remain red or inactive.
- Pings fail even though addressing appears correct.

Check:

```
show ip interface brief
show interfaces status
```

Use the following command in the appropriate interface configuration mode:

```
no shutdown
```

Grading Overview

This skills check is worth **100 points**.

Category	Points
Physical topology and device placement	10
VLAN creation and naming	10
Access-port configuration	10
Trunk configuration	15
Router-on-a-stick configuration	15
IPv4 addressing and PC configuration	15
Connectivity testing and verification	10
Professional documentation	10
Troubleshooting log and technical explanation	5
Total	100

A detailed rubric follows.

Skills Check Rubric

Category	Exemplary	Proficient	Developing	Beginning	Points
Physical Topology and Device Placement	All required devices are present, correctly connected, clearly labeled, and use appropriate interfaces and cables.	All required devices are present and connected with one minor labeling or cabling issue.	One or more devices, connections, or labels are missing or incorrect.	The topology is incomplete or does not represent the required network.	5
VLAN Creation and Naming	VLANs 10 and 20 exist on both switches with exact names and are verified using show commands.	VLANs exist on both switches with one minor naming or verification issue.	One VLAN is missing, incorrectly named, or configured on only one switch.	VLAN configuration is largely missing or incorrect.	5
Access-Port Configuration	All four PC ports are in access mode and assigned to the correct VLAN; interfaces are enabled.	Most access ports are correct with one minor error.	Multiple access-port assignments or modes are incorrect.	Access ports are not configured according to the requirements.	5
Trunk Configuration	Both required trunk links are operational, use trunk mode, and carry VLANs 10 and 20. Verification output is included.	Both trunk links are configured with one minor omission or verification issue.	One trunk is incorrect, inactive, or missing.	Trunk configuration is largely missing or prevents VLAN communication.	10
Router-on-a-Stick Configuration	Both subinterfaces use correct VLAN encapsulation, IP addresses, subnet masks, and an enabled physical interface.	Router subinterfaces are mostly correct with one minor error.	One subinterface is missing or incorrectly configured.	Router-on-a-stick configuration is absent or nonfunctional.	10
IPv4 Addressing and PC Configuration	All PCs have correct IP addresses, subnet masks, and default gateways based on the assigned subnets.	Most addressing is correct with one minor error.	Multiple address, mask, or gateway errors are present.	Addressing is incomplete or prevents meaningful testing.	10

Category	Exemplary	Proficient	Developing	Beginning	Points
Connectivity Testing and Verification	All required ping tests succeed, show commands are complete, and results are accurately recorded.	Most required tests succeed and verification is mostly complete.	Several tests fail or documentation of results is incomplete.	Testing is missing or results are not reliable.	5
Professional Documentation	Documentation is complete, organized, readable, technically accurate, and includes all required components and screenshots.	Documentation is mostly complete with minor organization or formatting issues.	Several required components are incomplete, unclear, or difficult to follow.	Documentation is substantially incomplete or unprofessional.	5
Troubleshooting Log and Technical Explanation	Troubleshooting is accurately documented, and the student clearly explains VLAN separation, trunking, default gateways, and inter-VLAN routing.	Explanation and troubleshooting log are generally accurate with minor omissions.	Explanation is incomplete or includes multiple technical errors.	Troubleshooting log and explanation are missing or inaccurate.	5
Total					60

Performance Levels

Score (Percent)	Performance Level
90–100	Advanced mastery
80–89	Proficient
70–79	Developing
Below 70	Beginning; additional practice recommended